

Exercise of Supervised Learning: Gaussian Processes 1

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Exercise 1: Bayesian Linear Model

Model:

$$y = f(\mathbf{x}) + \epsilon = \boldsymbol{\theta}^\top \mathbf{x} + \epsilon,$$

with $\epsilon \sim \mathcal{N}(0, \sigma^2) \perp \mathbf{x}$. Per data point:

$$y^{(i)} = \boldsymbol{\theta}^\top \mathbf{x}^{(i)} + \epsilon^{(i)}, \quad \epsilon^{(i)} \sim \mathcal{N}(0, \sigma^2) \text{ i.i.d., } i \in [n].$$

Bayesian view: $\boldsymbol{\theta}$ is random with a prior.

MAP estimate: $\hat{\boldsymbol{\theta}} = \arg \max_{\boldsymbol{\theta}} p(\boldsymbol{\theta} | \mathbf{X}, \mathbf{y})$.

(a) Uniform prior $q(\boldsymbol{\theta}) \propto 1 \Rightarrow$ show MAP = **L2-loss ERM** (OLS) over linear models.

Exercise 1: The Key Insight — MAP = Regularized ERM

Take $-\log$ of the posterior: maximizing $p(\theta|\mathbf{X}, \mathbf{y})$ becomes minimizing

$$\hat{\theta} = \arg \max_{\theta} p(\theta|\mathbf{X}, \mathbf{y}) = \arg \min_{\theta} \left[\underbrace{-\log p(\mathbf{y}|\mathbf{X}, \theta)}_{\text{data fit (L2 loss)}} + \underbrace{(-\log q(\theta))}_{\text{penalty}} \right].$$

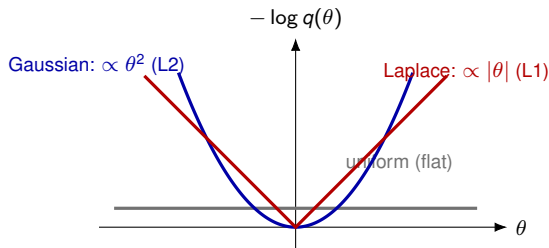
$p(\theta|\mathbf{X}, \mathbf{y}) \propto p(\mathbf{y}|\mathbf{X}, \theta) p(\theta)$

- ▶ The likelihood is always Gaussian $\leadsto$ data-fit term is the **L2 loss**.
- ▶ The **prior choice** $q(\theta)$ **determines the penalty**. Parts (a)/(b)/(c) just swap the prior:

Prior $q(\theta)$	$-\log q(\theta)$ (penalty)	Estimator	λ
(a) Uniform	const (none)	OLS / L2-loss ERM	—
(b) Gaussian	$\frac{1}{2\tau^2} \ \theta\ _2^2$	Ridge	σ^2/τ^2
(c) Laplace	$\frac{1}{\tau} \ \theta\ _1$	Lasso	$2\sigma^2/\tau$

Exercise 1: Why the Prior Is the Regularizer

The penalty $-\log q(\theta)$ as a function of a scalar parameter θ — its *shape* is what each prior contributes:



- **Flat** $\Rightarrow$ no shrinkage (OLS). **Parabola** $\Rightarrow$ smooth shrinkage (Ridge).
- **V-shape** has a kink at 0 $\Rightarrow$ drives coefficients exactly to 0 (Lasso, sparsity).

Exercise 1: Think Before Solving

Target: turn the $\arg \max$ of the posterior into a familiar $\arg \min$ (an ERM).

- ▶ **From max to min.** $\log$ is monotone — what does $-\log p(\theta|\mathbf{X}, \mathbf{y})$ do to the $\arg \max$? How does Bayes split it into likelihood $\times$ prior?
- ▶ **Data fit.** The likelihood $\mathbf{y}|\mathbf{X}, \theta$ is Gaussian. Which familiar loss does $-\log p(\mathbf{y}|\mathbf{X}, \theta)$ reduce to?
- ▶ **Penalty.** For each prior (uniform / Gaussian / Laplace), what does $-\log q(\theta)$ become, and why does the marginal $p(\mathbf{y}|\mathbf{X})$ never matter?

$$\arg \max p(\theta|\mathbf{X}, \mathbf{y}) \rightarrow \arg \min [-\log \text{lik.} - \log \text{prior}] = (\text{L2 loss}) + (\text{penalty})$$

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1 (a): Construct Posterior from Bayes' Rule

$$\underbrace{p(\theta|\mathbf{X}, \mathbf{y})}_{\text{posterior}} = \frac{p(\theta, \mathbf{y}|\mathbf{X})}{p(\mathbf{y}|\mathbf{X})}$$

- ▶ Linear model: $\mathbf{y}|\mathbf{X}, \theta \sim \mathcal{N}(\mathbf{X}\theta, \sigma^2 \mathbf{I})$ (next slide).
- ▶ **Uniform prior:** $q(\theta) \propto 1$.
- ▶ Marginal $p(\mathbf{y}|\mathbf{X}) \perp \theta \rightsquigarrow$ constant for the arg max.

$$\begin{aligned} p(\theta|\mathbf{X}, \mathbf{y}) &= \frac{p(\theta, \mathbf{X}, \mathbf{y})}{p(\mathbf{X}, \mathbf{y})} \\ &= \frac{p(\mathbf{y}|\mathbf{X}, \theta) p(\mathbf{X}, \theta)}{p(\mathbf{X}, \mathbf{y})} \\ &= \frac{p(\mathbf{y}|\mathbf{X}, \theta) p(\mathbf{X}) p(\theta)}{p(\mathbf{X}, \mathbf{y})} \\ &= \frac{p(\mathbf{y}|\mathbf{X}, \theta) p(\theta)}{p(\mathbf{y}|\mathbf{X})} \end{aligned}$$

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$$\underbrace{p(\theta|\mathbf{X}, \mathbf{y})}_{\text{posterior}} = \frac{p(\theta, \mathbf{y}|\mathbf{X})}{p(\mathbf{y}|\mathbf{X})}$$

$\max \log p(\theta|\mathbf{X}, \mathbf{y})$
 $= \max \log p(\mathbf{y}|\mathbf{X}, \theta) + \log q(\theta)$

$\underbrace{p(\mathbf{y}|\mathbf{X}, \theta)}_{\text{likelihood}} \underbrace{q(\theta)}_{\text{prior}}$
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1 (a): The Likelihood $p(\mathbf{y}|\mathbf{X}, \theta)$

$$\begin{aligned} p(\mathbf{y}|\mathbf{X}, \theta) &\propto \exp \left[-\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\theta)^\top (\mathbf{y} - \mathbf{X}\theta) \right] \\ &= \exp \left[-\frac{\|\mathbf{y} - \mathbf{X}\theta\|^2}{2\sigma^2} \right] \\ &= \exp \left[-\frac{\sum_{i=1}^n (y^{(i)} - \theta^\top \mathbf{x}^{(i)})^2}{2\sigma^2} \right]. \end{aligned}$$

Recall: prior $q(\theta) \propto 1$; marginal $p(\mathbf{y}|\mathbf{X})$ irrelevant.

$\rightsquigarrow p(\theta|\mathbf{X}, \mathbf{y}) \propto p(\mathbf{y}|\mathbf{X}, \theta) q(\theta)$.

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$\rightsquigarrow p(\theta|\mathbf{X}, \mathbf{y}) \propto p(\mathbf{y}|\mathbf{X}, \theta) q(\theta)$.

1 (a): MAP Estimate

$$p(\boldsymbol{\theta}|\mathbf{X}, \mathbf{y}) \propto p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) \cdot q(\boldsymbol{\theta}) \propto \exp \left[-\frac{\sum_{i=1}^n (y^{(i)} - \boldsymbol{\theta}^\top \mathbf{x}^{(i)})^2}{2\sigma^2} \right].$$

$$\begin{aligned}\hat{\boldsymbol{\theta}} &= \arg \max_{\boldsymbol{\theta}} p(\boldsymbol{\theta}|\mathbf{X}, \mathbf{y}) = \arg \max_{\boldsymbol{\theta}} \log p(\boldsymbol{\theta}|\mathbf{X}, \mathbf{y}) \\ &= \arg \max_{\boldsymbol{\theta}} -\frac{\sum_{i=1}^n (y^{(i)} - \boldsymbol{\theta}^\top \mathbf{x}^{(i)})^2}{2\sigma^2} = \arg \min_{\boldsymbol{\theta}} \frac{\sum_{i=1}^n (y^{(i)} - \boldsymbol{\theta}^\top \mathbf{x}^{(i)})^2}{2\sigma^2} \\ &= \arg \min_{\boldsymbol{\theta}} \sum_{i=1}^n (y^{(i)} - \boldsymbol{\theta}^\top \mathbf{x}^{(i)})^2. \quad \square\end{aligned}$$

$\Rightarrow \text{MAP} \Leftrightarrow \text{L2-loss ERM (OLS)}.$

Roadmap (Ex. 1)

► (a) Uniform

no penalty $\rightarrow$ OLS

(b) Gaussian

L2 penalty $\rightarrow$ Ridge

(c) Laplace

L1 penalty $\rightarrow$ Lasso

1 (a): MAP Estimate

$$p(\theta|\mathbf{X}, \mathbf{y}) \propto p(\mathbf{y}|\mathbf{X}, \theta) \cdot q(\theta) \propto \exp \left[-\frac{\sum_{i=1}^n (y^{(i)} - \theta^\top \mathbf{x}^{(i)})^2}{2\sigma^2} \right].$$

$$\hat{\theta} = \arg \max_{\theta} p(\theta|\mathbf{X}, \mathbf{y}) = \arg \max_{\theta} \log p(\theta|\mathbf{X}, \mathbf{y})$$

$$= \arg \max_{\theta} \left(-\frac{\sum_{i=1}^n (y^{(i)} - \theta^\top \mathbf{x}^{(i)})^2}{2\sigma^2} \right) = \arg \min_{\theta} \frac{\sum_{i=1}^n (y^{(i)} - \theta^\top \mathbf{x}^{(i)})^2}{2\sigma^2}$$

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$\Rightarrow \text{MAP} \Leftrightarrow \text{L2-loss ERM (OLS)}.$

Roadmap (Ex. 1)

► (a) Uniform

no penalty $\rightarrow$ OLS

(b) Gaussian

L2 penalty $\rightarrow$ Ridge

(c) Laplace

L1 penalty $\rightarrow$ Lasso

Exercise 1(b)

Gaussian prior:

$$q(\boldsymbol{\theta}) \propto \exp \left[-\frac{1}{2\tau^2} \boldsymbol{\theta}^\top \boldsymbol{\theta} \right], \quad \tau > 0,$$

Show: for a suitable τ , MAP = **Ridge** (L2-loss + L2 penalty) over linear models.

1 (b): Posterior with a Gaussian Prior

$$\begin{aligned} p(\boldsymbol{\theta}|\mathbf{X}, \mathbf{y}) &\propto p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta})q(\boldsymbol{\theta}) \\ &\propto \exp \left[-\frac{\sum_{i=1}^n (y^{(i)} - \boldsymbol{\theta}^\top \mathbf{x}^{(i)})^2}{2\sigma^2} - \frac{1}{2\tau^2} \boldsymbol{\theta}^\top \boldsymbol{\theta} \right] \\ &\propto \exp \left[-\frac{\sum_{i=1}^n (y^{(i)} - \boldsymbol{\theta}^\top \mathbf{x}^{(i)})^2}{2\sigma^2} - \frac{\|\boldsymbol{\theta}\|_2^2}{2\tau^2} \right]. \end{aligned}$$

Next: $\arg \max_{\boldsymbol{\theta}} p(\boldsymbol{\theta}|\mathbf{X}, \mathbf{y}) = \arg \max_{\boldsymbol{\theta}} \log p(\boldsymbol{\theta}|\mathbf{X}, \mathbf{y})$.

1 (b): MAP Estimate

$$\begin{aligned}\hat{\theta} &= \arg \max_{\theta} \log p(\theta | \mathbf{X}, \mathbf{y}) \\&= \arg \max_{\theta} - \frac{\sum_{i=1}^n (y^{(i)} - \theta^{\top} \mathbf{x}^{(i)})^2}{2\sigma^2} - \frac{\|\theta\|_2^2}{2\tau^2} \\&= \arg \min_{\theta} \left(\frac{\sum_{i=1}^n (y^{(i)} - \theta^{\top} \mathbf{x}^{(i)})^2}{2\sigma^2} + \frac{\|\theta\|_2^2}{2\tau^2} \right) \\&= \arg \min_{\theta} \sum_{i=1}^n (y^{(i)} - \theta^{\top} \mathbf{x}^{(i)})^2 + \frac{\sigma^2}{\tau^2} \|\theta\|_2^2. \quad \square\end{aligned}$$

(Because $\times 2\sigma^2$ doesn't change the argmin.)

Set $\lambda = \frac{\sigma^2}{\tau^2} \Rightarrow \text{MAP} \Leftrightarrow \text{L2-loss} + \text{L2 penalty (Ridge)}.$

Roadmap (Ex. 1)

✓ (a) Uniform

no penalty $\rightarrow$ OLS

► (b) Gaussian

L2 penalty $\rightarrow$ Ridge

(c) Laplace

L1 penalty $\rightarrow$ Lasso

Exercise 1(c)

Laplace prior:


$$q(\boldsymbol{\theta}) \propto \exp \left[-\frac{\sum_{j=1}^p |\theta_j|}{\tau} \right], \quad \tau > 0,$$

Show: for a suitable τ , MAP = **Lasso** (L2-loss + L1 penalty) over linear models.

1 (c): Posterior with a Laplace Prior

$$\begin{aligned} p(\boldsymbol{\theta}|\mathbf{X}, \mathbf{y}) &\propto p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta})q(\boldsymbol{\theta}) \\ &\propto \exp \left[-\frac{\sum_{i=1}^n (y^{(i)} - \boldsymbol{\theta}^\top \mathbf{x}^{(i)})^2}{2\sigma^2} - \frac{\|\boldsymbol{\theta}\|_1}{\tau} \right]. \end{aligned}$$

1 (c): MAP Estimate

$$\begin{aligned}\hat{\theta} &= \arg \max_{\theta} \log p(\theta | \mathbf{X}, \mathbf{y}) \\ &= \arg \max_{\theta} - \frac{\sum_{i=1}^n (y^{(i)} - \theta^{\top} \mathbf{x}^{(i)})^2}{2\sigma^2} - \frac{\|\theta\|_1}{\tau} \\ &= \arg \min_{\theta} \left[\frac{\sum_{i=1}^n (y^{(i)} - \theta^{\top} \mathbf{x}^{(i)})^2}{2\sigma^2} + \frac{\|\theta\|_1}{\tau} \right] \quad \times 2\sigma^2 \\ &= \arg \min_{\theta} \sum_{i=1}^n (y^{(i)} - \theta^{\top} \mathbf{x}^{(i)})^2 + \frac{2\sigma^2}{\tau} \|\theta\|_1. \quad \square\end{aligned}$$

(Because $\times 2\sigma^2$ doesn't change the argmin.)

Set $\lambda = \frac{2\sigma^2}{\tau} \Rightarrow \text{MAP} \Leftrightarrow \text{L2-loss} + \text{L1 penalty (Lasso)}.$

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no penalty $\rightarrow$ OLS

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Exercise 2: Covariance Functions

Covariance functions from the lecture: constant, linear, polynomial, squared-exp, Matérn, exponential.

(a) Show each is *valid*. (Matérn & exp. omitted.) Use the composition rules (next slide).

(b) Which are stationary / isotropic? Justify.

2 (a): Toolbox — Composition Rules

Assume k_0, k_1 are valid covariance functions. Proofs below cite these:

1. $k(\mathbf{x}, \mathbf{x}') = \mathbf{x}^\top \mathbf{x}'$;

2. $k(\mathbf{x}, \mathbf{x}') = c \cdot k_0(\mathbf{x}, \mathbf{x}')$, $c \geq 0$ const;

3. $k(\mathbf{x}, \mathbf{x}') = k_0(\mathbf{x}, \mathbf{x}') + k_1(\mathbf{x}, \mathbf{x}')$;

4. $k(\mathbf{x}, \mathbf{x}') = k_0(\mathbf{x}, \mathbf{x}') \cdot k_1(\mathbf{x}, \mathbf{x}')$;

5. $k(\mathbf{x}, \mathbf{x}') = g(k_0(\mathbf{x}, \mathbf{x}'))$, g polynomial, **pos.** coeff.;

6. $k(\mathbf{x}, \mathbf{x}') = t(\mathbf{x}) \cdot k_0(\mathbf{x}, \mathbf{x}') \cdot t(\mathbf{x}')$, t **continuous**;

7. $k(\mathbf{x}, \mathbf{x}') = \exp(k_1(\mathbf{x}, \mathbf{x}'))$;

8. $k(\mathbf{x}, \mathbf{x}') = \mathbf{x}^\top \mathbf{A} \mathbf{x}'$, $\mathbf{A} \succeq 0$.

2 (a): Think Before Solving

Target: show each covariance function is *valid* (a positive semi-definite kernel).

- ▶ **Two routes (single kernel).** (i) Gram matrix $\mathbf{K}$ symmetric with $\mathbf{v}^\top \mathbf{K} \mathbf{v} \geq 0 \ \forall \mathbf{v}$, or (ii) $k(\mathbf{x}, \mathbf{x}') = \langle \phi(\mathbf{x}), \phi(\mathbf{x}') \rangle$. Which is easier for the constant kernel?
- ▶ **Build, don't re-derive.** Can you assemble composite kernels from already-valid pieces via the toolbox rules?
- ▶ **Decomposition.** Linear = constant + inner product; polynomial = $(\cdot)^P$; squared-exp = $t \cdot k_4 \cdot t$. Which rule fits each step?

PSD Gram matrix or feature-map inner product then compose via rules 1–8

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2 (a): Constant Kernel — Proof via Kernel Matrix

Kernel (Gram) matrix $\mathbf{K} \in \mathbb{R}^{n \times n}$ on $\mathcal{D} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\}$, $K_{i,j} = k(\mathbf{x}, \mathbf{x}')$.

Here $k(\mathbf{x}, \mathbf{x}') \equiv \sigma_0^2$, so $K_{i,j} = \sigma_0^2$:

$$\mathbf{K} = \begin{pmatrix} \sigma_0^2 & \dots & \sigma_0^2 \\ \vdots & \ddots & \vdots \\ \sigma_0^2 & \dots & \sigma_0^2 \end{pmatrix}$$

Note: $\mathbf{K}$ (matrix) $\neq k(\cdot, \cdot)$ (function) — never write “ k is P.S.D.” in the exam.

Prove $\mathbf{K}$ is P.S.D.:

1. Prove $\mathbf{K}$ is symmetric.
2. Prove that $\forall \mathbf{v} \in \mathbb{R}^n$, $\mathbf{v}^\top \mathbf{K} \mathbf{v} \geq 0$.

Roadmap (Ex. 2a)

► Constant

σ_0^2 : Gram PSD / $\phi = \sigma_0$

Linear

const + inner prod.

Polynomial

$(\cdot)^p$, pos. coeff.

Squared exp.

$t \cdot K_4 \cdot t$

$$K(\mathbf{x}, \mathbf{x}') = \phi(\mathbf{x})^\top \phi(\mathbf{x}') = \sigma_0^2$$
$$\phi(\mathbf{x}) = \sigma_0$$

2 (a): How to Prove That A Matrix Is P.S.D.

1. Since $K_{i,j} = \sigma_0^2$ for all i, j , we have $\mathbf{K}^\top = \mathbf{K}$, thus $\mathbf{K}$ is symmetric.
2. For any $\mathbf{v} = (v_1, \dots, v_n)^\top$: show $\mathbf{v}^\top \mathbf{K} \mathbf{v} \geq 0$.

2.1 Naive: compute $\mathbf{K} \mathbf{v}$, then $\mathbf{v}^\top (\mathbf{K} \mathbf{v})$.

$$\begin{aligned}\mathbf{K} \mathbf{v} &= \sigma_0^2 \left(\sum_i v_i, \sum_i v_i, \dots, \sum_i v_i \right)^\top, \\ \mathbf{v}^\top \mathbf{K} \mathbf{v} &= \sigma_0^2 [v_1(\sum_i v_i) + v_2(\sum_i v_i) + \dots + v_n(\sum_i v_i)] \\ &= \sigma_0^2 (v_1 + v_2 + \dots + v_n)(\sum_i v_i) = \sigma_0^2 \left(\sum_i v_i \right)^2 \geq 0.\end{aligned}$$

2.2 Faster way: $\mathbf{K} = \sigma_0^2 \mathbf{1} \mathbf{1}^\top$, where $\mathbf{1} = (1, 1, \dots, 1)^\top$. So
 $\mathbf{v}^\top \mathbf{K} \mathbf{v} = \sigma_0^2 \mathbf{v}^\top \mathbf{1} \mathbf{1}^\top \mathbf{v} = \sigma_0^2 (\mathbf{1}^\top \mathbf{v})^2 \geq 0$.

2 (a): How to Prove That A Matrix Is P.S.D.

1. Since $K_{i,j} = \sigma_0^2$ for all i, j , we have $\mathbf{K}^\top = \mathbf{K}$, thus $\mathbf{K}$ is symmetric.
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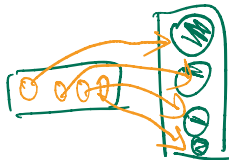
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$$\begin{aligned}\mathbf{K} \mathbf{v} &= \sigma_0^2 \left(\sum_i v_i, \sum_i v_i, \dots, \sum_i v_i \right)^\top, \\ \mathbf{v}^\top \mathbf{K} \mathbf{v} &= \sigma_0^2 \left[v_1 \left(\sum_i v_i \right) + v_2 \left(\sum_i v_i \right) + \dots + v_n \left(\sum_i v_i \right) \right] \\ &= \sigma_0^2 (v_1 + v_2 + \dots + v_n) \left(\sum_i v_i \right) = \sigma_0^2 \left(\sum_i v_i \right)^2 \geq 0.\end{aligned}$$

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2 (a): How to Prove That A Matrix Is P.S.D.

1. Since $K_{i,j} = \sigma_0^2$ for all i, j , we have $\mathbf{K}^\top = \mathbf{K}$, thus $\mathbf{K}$ is symmetric.
2. For any $\mathbf{v} = (v_1, \dots, v_n)^\top$: show $\mathbf{v}^\top \mathbf{K} \mathbf{v} \geq 0$.
 - 2.1 Naive: compute $\mathbf{K} \mathbf{v}$, then $\mathbf{v}^\top (\mathbf{K} \mathbf{v})$.



$$\begin{aligned}\mathbf{K} \mathbf{v} &= \sigma_0^2 (\underbrace{\sum_i v_i}_{\text{sum}}, \underbrace{\sum_i v_i}_{\text{sum}}, \dots, \underbrace{\sum_i v_i}_{\text{sum}})^\top, \\ \mathbf{v}^\top \mathbf{K} \mathbf{v} &= \sigma_0^2 [v_1(\sum_i v_i) + v_2(\sum_i v_i) + \dots + v_n(\sum_i v_i)] \\ &= \sigma_0^2 (v_1 + v_2 + \dots + v_n)(\sum_i v_i) = \sigma_0^2 (\sum_i v_i)^2 \geq 0.\end{aligned}$$

- 2.2 Faster way: $\mathbf{K} = \sigma_0^2 \mathbf{1} \mathbf{1}^\top$, where $\mathbf{1} = (1, 1, \dots, 1)^\top$. So
 $\mathbf{v}^\top \mathbf{K} \mathbf{v} = \sigma_0^2 \mathbf{v}^\top \mathbf{1} \mathbf{1}^\top \mathbf{v} = \sigma_0^2 (\mathbf{1}^\top \mathbf{v})^2 \geq 0$.

$$\begin{aligned}\sigma_0^2 \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} \\ = \sigma_0^2 \begin{bmatrix} \boxed{\sum_i v_i} \\ \boxed{\sum_i v_i} \\ \boxed{\sum_i v_i} \\ \boxed{\sum_i v_i} \end{bmatrix} \\ \swarrow \\ \sum_i v_i\end{aligned}$$

2 (a): How to Prove That A Matrix Is P.S.D.

1. Since $K_{i,j} = \sigma_0^2$ for all i, j , we have $\mathbf{K}^\top = \mathbf{K}$, thus $\mathbf{K}$ is symmetric.
2. For any $\mathbf{v} = (v_1, \dots, v_n)^\top$: show $\mathbf{v}^\top \mathbf{K} \mathbf{v} \geq 0$.
 - 2.1 Naive: compute $\mathbf{K} \mathbf{v}$, then $\mathbf{v}^\top (\mathbf{K} \mathbf{v})$.

$$\begin{aligned}\mathbf{K} \mathbf{v} &= \sigma_0^2 \left(\sum_i v_i, \sum_i v_i, \dots, \sum_i v_i \right)^\top, \\ \mathbf{v}^\top \mathbf{K} \mathbf{v} &= \sigma_0^2 \left[v_1 \left(\sum_i v_i \right) + v_2 \left(\sum_i v_i \right) + \dots + v_n \left(\sum_i v_i \right) \right] \\ &= \sigma_0^2 (v_1 + v_2 + \dots + v_n) \left(\sum_i v_i \right) = \sigma_0^2 \left(\sum_i v_i \right)^2 \geq 0.\end{aligned}$$

- 2.2 Faster way: $\mathbf{K} = \sigma_0^2 \mathbf{1} \mathbf{1}^\top$, where $\mathbf{1} = (1, 1, \dots, 1)^\top$. So
 $\mathbf{v}^\top \mathbf{K} \mathbf{v} = \sigma_0^2 \mathbf{v}^\top \mathbf{1} \mathbf{1}^\top \mathbf{v} = \sigma_0^2 (\mathbf{1}^\top \mathbf{v})^2 \geq 0$.

$$\mathbf{1}^\top \mathbf{v} = \mathbf{v}^\top \mathbf{1} \in \mathbb{R}$$

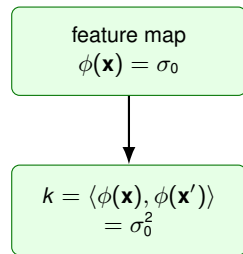


2 (a): Constant Kernel — Proof via Feature Map

Alternative: write $k(\mathbf{x}, \mathbf{x}') = \sigma_0^2$ as a **feature-map inner product** $\langle \phi(\mathbf{x}), \phi(\mathbf{x}') \rangle$.

$\Rightarrow$ construct $\phi(\mathbf{x}) \in \mathbb{R}^d$ explicitly. Here $\phi(\mathbf{x}) = \sigma_0$:

$$k(\mathbf{x}, \mathbf{x}') = \sigma_0^2 = \langle \sigma_0, \sigma_0 \rangle = \langle \phi(\mathbf{x}), \phi(\mathbf{x}') \rangle. \quad \square$$

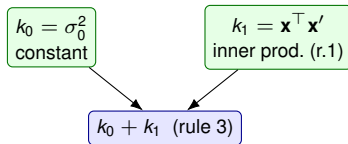


An inner product is always a valid kernel.

2 (a): Linear Kernel $k(\mathbf{x}, \mathbf{x}') = \sigma_0^2 + \mathbf{x}^\top \mathbf{x}'$

$$k(\mathbf{x}, \mathbf{x}') = \underbrace{\sigma_0^2}_{:=k_0(\mathbf{x}, \mathbf{x}')} + \underbrace{\mathbf{x}^\top \mathbf{x}'}_{:=k_1(\mathbf{x}, \mathbf{x}')}$$

1. We have shown that k_0 is a valid cov. func.
2. k_1 is an inner product $\langle \phi(\mathbf{x}), \phi(\mathbf{x}') \rangle$ with $\phi(\mathbf{x}) := \mathbf{x}$. So k_1 is a valid cov. func. (rule 1).
3. Their sum $k_0 + k_1$ is also a cov. func. (rule 3). □



Roadmap (Ex. 2a)

✓ **Constant**

σ_0^2 : Gram PSD / $\phi = \sigma_0$

► **Linear**

const + inner prod.

Polynomial

$(\cdot)^p$, pos. coeff.

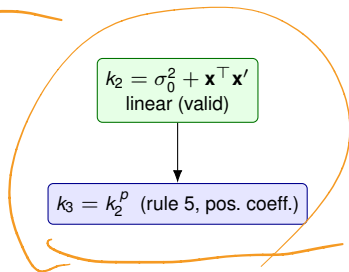
Squared exp.

$t \cdot k_4 \cdot t$

2 (a): Polynomial Kernel $k(\mathbf{x}, \mathbf{x}') = (\underbrace{\sigma_0^2 + \mathbf{x}^\top \mathbf{x}'}_{})^p$

We define $k_2(\mathbf{x}, \mathbf{x}') = \sigma_0^2 + \mathbf{x}^\top \mathbf{x}'$, so $k_3 = (\sigma_0^2 + \mathbf{x}^\top \mathbf{x}')^p = k_2^p$.

1. We have shown that k_2 (the linear kernel) is a cov. func.
2. $k_3 = k_2^p$: a polynomial in k_2 , single term with **positive coefficient**
 $1 \Rightarrow$ valid (rule 5). □



Roadmap (Ex. 2a)

✓ **Constant**

σ_0^2 : Gram PSD / $\phi = \sigma_0$

✓ **Linear**

const + inner prod.

► **Polynomial**

$(\cdot)^p$, pos. coeff.

Squared exp.

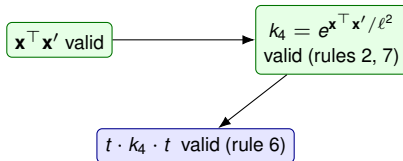
$t \cdot k_4 \cdot t$

2 (a): Squared-Exp. Kernel $k(\mathbf{x}, \mathbf{x}') = \exp(-\|\mathbf{x} - \mathbf{x}'\|_2^2 / 2\ell^2)$

$$\begin{aligned}
 k(\mathbf{x}, \mathbf{x}') &= \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|_2^2}{2\ell^2}\right) \\
 &= \exp\left(-\frac{\mathbf{x}^\top \mathbf{x} - 2\mathbf{x}^\top \mathbf{x}' + \mathbf{x}'^\top \mathbf{x}'}{2\ell^2}\right) \\
 &= \underbrace{\exp\left(-\frac{\mathbf{x}^\top \mathbf{x}}{2\ell^2}\right)}_{:=t(\mathbf{x})} \cdot \underbrace{\exp\left(\frac{\mathbf{x}^\top \mathbf{x}'}{\ell^2}\right)}_{:=k_4(\mathbf{x}, \mathbf{x}')} \cdot \underbrace{\exp\left(-\frac{\mathbf{x}'^\top \mathbf{x}'}{2\ell^2}\right)}_{:=t(\mathbf{x}')},
 \end{aligned}$$

$\exp(a+b+c)$
 $= \exp(a) \cdot \exp(b) \cdot \exp(c)$

with a continuous function $t(\cdot)$. Now build up the factors:



Roadmap (Ex. 2a)

✓ **Constant**

σ_0^2 : Gram PSD / $\phi = \sigma_0$

✓ **Linear**

const + inner prod.

✓ **Polynomial**

$(\cdot)^p$, pos. coeff.

► **Squared exp.**

$t \cdot k_4 \cdot t$

2 (b): Stationary vs. Isotropic — Overview

(a, b)
 (c, d)

Definitions. k is **stationary** if $k(\mathbf{x}, \mathbf{x} + \mathbf{d}) = k(\mathbf{0}, \mathbf{d})$ (depends only on the shift $\mathbf{d}$).
 k is **isotropic** if it is a function of $\|\mathbf{x} - \mathbf{x}'\|$ (depends only on the distance).

Cov. function	Stationary?	Isotropic?	Reason
Constant σ_0^2	✓	✓	const = $\sigma_0^2 \ \mathbf{x} - \mathbf{x}'\ ^0$
Linear $\sigma_0^2 + \mathbf{x}^\top \mathbf{x}'$	×	×	has $\mathbf{x}^\top \mathbf{d}$ term
Polynomial $(\sigma_0^2 + \mathbf{x}^\top \mathbf{x}')^p$	×	×	same as linear
Squared exp.	✓	✓	function of $\ \mathbf{x} - \mathbf{x}'\ $
Matérn	✓	✓	function of $\ \mathbf{x} - \mathbf{x}'\ $
Exponential	✓	✓	function of $\ \mathbf{x} - \mathbf{x}'\ $

if $b-a = d-c$
then
 $k(a, b) = k(c, d)$

2 (b): Constant functions

1. $k(\mathbf{x}, \mathbf{x}') = \sigma_0^2$ is stationary since $k(\mathbf{x}, \mathbf{x} + \mathbf{d}) = \sigma_0^2 = k(\mathbf{0}, \mathbf{d})$.

2. $k(\mathbf{x}, \mathbf{x}') = \sigma_0^2$ is isotropic since $k(\mathbf{x}, \mathbf{x}') = \sigma_0^2 \|\mathbf{x} - \mathbf{x}'\|^0$.

2 (b): Constant functions

1. $k(\mathbf{x}, \mathbf{x}') = \sigma_0^2$ is stationary since $k(\mathbf{x}, \mathbf{x} + \mathbf{d}) = \sigma_0^2 = k(\mathbf{0}, \mathbf{d})$.
2. $k(\mathbf{x}, \mathbf{x}') = \sigma_0^2$ is isotropic since $k(\mathbf{x}, \mathbf{x}') = \sigma_0^2 \|\mathbf{x} - \mathbf{x}'\|^0$.

2 (b): $k(\mathbf{x}, \mathbf{x}') = \sigma_0^2 + \mathbf{x}^\top \mathbf{x}'$

1. $k(\mathbf{x}, \mathbf{x}') = \sigma_0^2 + \mathbf{x}^\top \mathbf{x}'$ is NOT stationary, since

$$\begin{aligned} k(\mathbf{x}, \mathbf{x} + \mathbf{d}) &= \sigma_0^2 + \mathbf{x}^\top \mathbf{x} + \mathbf{x}^\top \mathbf{d} \\ k(\mathbf{0}, \mathbf{d}) &= \sigma_0^2. \end{aligned}$$

$$k(a, b) \neq k(c, d)$$

$$b - a = d - c$$

2. It is NOT isotropic, since it cannot be written as a func. of $\|\mathbf{x} - \mathbf{x}'\|$.

The polynomial case is analogous — neither stationary nor isotropic.

2 (b): $k(\mathbf{x}, \mathbf{x}') = \sigma_0^2 + \mathbf{x}^\top \mathbf{x}'$

1. $k(\mathbf{x}, \mathbf{x}') = \sigma_0^2 + \mathbf{x}^\top \mathbf{x}'$ is NOT stationary, since

$$k(\mathbf{x}, \mathbf{x} + \mathbf{d}) = \sigma_0^2 + \mathbf{x}^\top \mathbf{x} + \mathbf{x}^\top \mathbf{d}$$

$$k(\mathbf{0}, \mathbf{d}) = \sigma_0^2.$$

2. It is NOT isotropic, since it cannot be written as a func. of $\|\mathbf{x} - \mathbf{x}'\|$.

The polynomial case is analogous — neither stationary nor isotropic.

2 (b): $k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|_2^2}{2\ell^2}\right)$

$\mathbf{x} - \mathbf{x}'$

1. $k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|_2^2}{2\ell^2}\right)$ is stationary, since
 $k(\mathbf{x}, \mathbf{x} + \mathbf{d}) = \exp\left(-\frac{\|\mathbf{d}\|_2^2}{2\ell^2}\right) = k(\mathbf{0}, \mathbf{d})$.
2. It is isotropic, since it is a function of $\|\mathbf{x} - \mathbf{x}'\|$.

2 (b): $k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|_2^2}{2\ell^2}\right)$

1. $k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|_2^2}{2\ell^2}\right)$ is stationary, since
 $k(\mathbf{x}, \mathbf{x} + \mathbf{d}) = \exp\left(-\frac{\|\mathbf{d}\|_2^2}{2\ell^2}\right) = k(\mathbf{0}, \mathbf{d})$.
2. It is isotropic, since it is a function of $\|\mathbf{x} - \mathbf{x}'\|$.

2 (b): Matérn and Exponential Cov. Func.

Same argument as the squared-exp kernel — both are functions of $\|\mathbf{x} - \mathbf{x}'\|$, hence **stationary & isotropic**:

1. Matérn:

$$k(\mathbf{x}, \mathbf{x}') = \frac{1}{2^\nu} \left(\frac{\sqrt{2\nu}}{\ell} \|\mathbf{x} - \mathbf{x}'\| \right)^\nu K_\nu \left(\frac{\sqrt{2\nu}}{\ell} \|\mathbf{x} - \mathbf{x}'\| \right)$$

2. Exponential:

$$k(\mathbf{x}, \mathbf{x}') = \exp \left(-\frac{\|\mathbf{x} - \mathbf{x}'\|}{\ell} \right)$$

Summary — Connecting the exercise to the lecture

	What we derived	Lecture concept made concrete
Ex. 1	$\text{MAP} = \arg \min[-\log \text{lik} - \log \text{prior}]$: the prior is the regularizer. Uniform $\rightarrow$ OLS, Gaussian $\rightarrow$ Ridge ($\lambda = \sigma^2/\tau^2$), Laplace $\rightarrow$ Lasso ($\lambda = 2\sigma^2/\tau$).	MAP estimation as regularized ERM; L2 loss from a Gaussian likelihood; L2/L1 penalties from the prior.
Ex. 2(a)	Proved validity of constant, linear, polynomial and squared-exp. kernels via a PSD Gram matrix, a feature-map inner product, and the composition rules.	Valid covariance functions; PSD kernel matrices, feature maps, and building kernels from simpler ones.
Ex. 2(b)	Constant / squared-exp. / Matérn / exponential are stationary & isotropic; linear & polynomial are neither.	Stationarity (depends on the shift) vs. isotropy (depends on the distance).